

EXPERIMENT 2

Bias Audit on Pre-trained Word Embeddings using WEAT

AIM:

To develop a program to perform Bias Audit on pre-trained word embeddings to identify gender or racial stereotypes using the Word Embedding Association Test (WEAT) on GloVe (Common Crawl) embeddings.

PROCEDURE:

1. Install required libraries: gensim, scipy, numpy.
2. Download the GloVe 840B 300d pre-trained word embeddings from Stanford NLP.
3. Load embeddings into a dictionary mapping words to 300-dimensional numpy float32 vectors.
4. Define cosine_similarity(vec1, vec2) using scipy's cosine distance function.
5. Define association(w, A, B, embeddings): compute mean cosine similarity of word w to target sets A and B.
6. Define weat_effect_size(X, Y, A, B, embeddings): calculate WEAT effect size as $(\text{mean_s_X} - \text{mean_s_Y}) / \text{std}(\text{s_X} + \text{s_Y})$.
7. Filter word sets to retain only words present in the embeddings vocabulary.
8. Gender Bias Test: X = male words, Y = female words, A = career terms, B = family terms.
9. Compute effect size and interpret: positive value $\rightarrow$ male words more associated with career.
10. Racial Bias Test: X = traditionally White names (john, paul, mike), Y = Black names (lakisha, tanisha, latoya), A = pleasant words, B = unpleasant words.
11. Print effect sizes and interpretations for both bias audits.

CODE:

```
!pip install gensim scipy numpy
!wget http://nlp.stanford.edu/data/glove.840B.300d.zip
!unzip glove.840B.300d.zip

import numpy as np
from scipy.spatial.distance import cosine

embedding_file = 'glove.840B.300d.txt'
embeddings = {}
with open(embedding_file, 'r', encoding='utf-8') as f:
    for line in f:
        values = line.split()
        if len(values) < 2: continue
        word = values[0]
        try:
            vector = np.asarray(values[1:], dtype='float32')
            embeddings[word] = vector
        except ValueError: continue

def cosine_similarity(vec1, vec2):
    return 1 - cosine(vec1, vec2)

def association(w, A, B, embeddings):
    return np.mean([cosine_similarity(embeddings[w], embeddings[a]) for a in A]) - \
           np.mean([cosine_similarity(embeddings[w], embeddings[b]) for b in B])

def weat_effect_size(X, Y, A, B, embeddings):
    s_X = [association(x, A, B, embeddings) for x in X]
    s_Y = [association(y, A, B, embeddings) for y in Y]
    numerator = np.mean(s_X) - np.mean(s_Y)
```

```
denominator = np.std(s_X + s_Y)
return numerator / denominator

X = ['man', 'male', 'boy', 'brother', 'he']
Y = ['woman', 'female', 'girl', 'sister', 'she']
A = ['career', 'profession', 'salary', 'office', 'business']
B = ['home', 'family', 'children', 'parents', 'marriage']

def filter_words(word_list, embeddings):
    return [w for w in word_list if w in embeddings]

X=filter_words(X,embeddings); Y=filter_words(Y,embeddings)
A=filter_words(A,embeddings); B=filter_words(B,embeddings)

effect_size = weat_effect_size(X, Y, A, B, embeddings)
print('WEAT Bias Audit Result')
print(f'Effect Size: {effect_size:.4f}')
if effect_size > 0:
    print('Interpretation: Male words are more associated with career terms.')

X2=['john', 'paul', 'mike']; Y2=['lakisha', 'tanisha', 'latoya']
A2=['pleasant', 'happy', 'joy', 'peace']; B2=['ugly', 'evil', 'pain', 'terrible']
X2=filter_words(X2,embeddings); Y2=filter_words(Y2,embeddings)
A2=filter_words(A2,embeddings); B2=filter_words(B2,embeddings)
effect_size_race = weat_effect_size(X2, Y2, A2, B2, embeddings)
print('Racial Bias WEAT Result')
print(f'Effect Size: {effect_size_race:.4f}')
```

OUTPUT:

Filtered word sets:

X: ['man', 'male', 'boy', 'brother', 'he']

Y: ['woman', 'female', 'girl', 'sister', 'she']

A: ['career', 'profession', 'salary', 'office', 'business']

B: ['home', 'family', 'children', 'parents', 'marriage']

WEAT Bias Audit Result

Effect Size: 0.3619

Interpretation: Male words are more associated with career terms.

Racial Bias WEAT Result

Effect Size: 1.5544

RESULT:

The WEAT-based bias audit successfully revealed biases in GloVe word embeddings. The Gender Bias test yielded an effect size of 0.3619, confirming that male words are more associated with career-related terms. The Racial Bias test yielded an effect size of 1.5544, indicating significant association of traditionally White names with pleasant attributes compared to Black names, exposing racial stereotypes in pre-trained embeddings.